

Decoupling Language-Model Judgement from Constrained Portfolio Optimisation: A Measurement Protocol and Three Null Results

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Abstract—Language models are increasingly proposed as portfolio allocators. We argue that the interesting question is not whether such a model can produce weights, but whether it should, and we give an architecture and a measurement protocol that make the question answerable. The model’s only output is a bounded confidence score that tilts expected returns inside a convex program; the solver, which alone sees the covariance structure, the caps and the turnover penalty, produces every weight. To test what that separation is worth, we run four arms over thirteen years on the same eligible universe: the model with company names, the model with names removed, a deterministic factor control with no model, and a monolithic arm in which the model returns weights directly. All three effects are indistinguishable from zero: temporal contamination $+0.38$ pp/yr ($p = 0.91$), value added over the deterministic control -0.05 pp/yr ($p = 0.99$), and the value of decoupling $+0.81$ pp/yr ($p = 0.78$). The monolithic arm breached no ceiling, but returned weights that failed to sum to one in five of thirteen years and silently omitted eligible names in two. Because null results are only informative when the measuring instrument is sound, we describe the evaluation infrastructure in equal detail: a survivorship-free panel rebuilt from the exchange’s own quotation archive (514 issuers, 166 of which stop trading), receipt-date gating of regulatory filings, a nested selection protocol whose hindsight premium is measured at 0.65 pp/yr, and a deflated-Sharpe probability of 0.986 over 36 trials. We report seven defects found by auditing our own backtest, all of which had inflated the result.

Index Terms—portfolio optimisation, large language models, backtest overfitting, deflated Sharpe ratio, survivorship bias, temporally available data, emerging markets

I. Introduction

A language model asked for portfolio weights will produce them. Whether it should be asked is a separate question, and the literature has largely skipped it: papers that report a model beating a benchmark rarely include the control that would distinguish the model’s judgement from the ranking signal it was given, and rarely test whether the model profited from recognising companies whose subsequent fate is in its training corpus.

This paper takes the opposite route. We fix an architecture in which the model cannot set a weight, then measure what that restriction costs. The measurement requires three comparisons that are usually conflated:

- 1) Temporal contamination. Does the model do better when it can see company names than when it sees only anonymised financial facts? A positive, significant gap means it is remembering, not reasoning.

- 2) Added value. Does the model beat a deterministic control that ranks the same universe by the same pre-declared factor? Without this arm, any result merely restates the factor.
- 3) Value of decoupling. Does routing a bounded score through a convex solver beat letting the model emit weights directly? This is what the architecture is for.

Our contribution is threefold. First, the decoupled architecture and the protocol that makes these three questions separately answerable. Second, three null results on Brazilian equities over thirteen years, reported with the power caveats they deserve. Third, and inseparably, the evaluation infrastructure: a null result is only informative if the instrument would have detected an effect, so we document the panel construction, availability gating, the multiple-testing correction and the defects we found in our own work. Upon acceptance, we will deposit the evaluation artefacts and a versioned manifest in a permanent repository; during double-anonymous review, the same materials are available in an anonymised supplementary archive.

II. Related work

Markowitz’s mean-variance formulation [1] remains the allocation substrate, with estimation error, concentration and turnover treated as first-class constraints rather than afterthoughts. Black and Litterman [2] established the pattern we follow structurally: qualitative views are combined with a mechanical allocator through a defined interface rather than replacing it. DeMiguel et al. [3] show that optimised portfolios frequently fail to dominate naive diversification out of sample, which motivates explicit caps and a frozen comparator. Novy-Marx and Velikov [4] show that frictions can erase anomaly profitability, which is why cost and tax are inside the evaluated protocol here rather than appended to it.

On inference, Bailey and López de Prado [5] give the deflated Sharpe ratio, which corrects the observed statistic for the number of trials and the higher moments of the return distribution; Bailey et al. [6] give the probability of backtest overfitting. Harvey et al. [7] document how weak the standard significance bar is once the size of the search is acknowledged. These are not optional refinements for a paper that evaluates 36 configurations.

Financial language models have been studied as domain models, research assistants and direct stock selectors.

BloombergGPT demonstrates the value of finance-specific corpora for general financial tasks [8], while FinGPT argues for an open, data-centric pipeline [9]. Direct market applications report that news-conditioned language-model scores can predict returns [10] and that live ChatGPT ratings co-move with subsequent earnings and returns [11]. Those results motivate our experiment but do not answer its architectural question. A model may appear predictive because it was handed an effective structured signal, because names reveal outcomes embedded in its training data, or because invalid allocations are repaired off-record. Named, anonymised, deterministic and monolithic arms are required to distinguish these mechanisms.

III. Architecture

A. The interface

Let $\mu \in \mathbb{R}^n$ be the expected-return vector estimated from information available at the decision date, Σ the covariance estimate, and w_{prev} the incumbent book. The model returns a confidence vector $c \in [-1, +1]^n$, and the only channel through which it reaches the portfolio is

$$\mu_i \leftarrow \mu_i \cdot (1 + \lambda c_i), \quad c_i \in [-1, +1]. \quad (1)$$

The allocation is then the solution of

$$\max_w \mu^\top w - \frac{\gamma}{2} w^\top \Sigma w - \kappa \|w - w_{\text{prev}}\|_1 \quad (2)$$

subject to $\sum_i w_i = 1$, $w \geq 0$, an asset-class ceiling, and a per-issuer ceiling. The constants λ , γ and κ are fixed before the run and are not tuned per period.

One clarification matters for reproducibility, because the two are easy to conflate. The solver above is the allocator for every arm in the language-model experiment. The eleven-year published strategy uses a simpler allocator on the same screen: a hard top- N cut on the factor score, one share class per issuer, with weights assigned in proportion to score inside the same ceilings. We report both because the paper’s claims about the model concern the first and the claims about realised returns concern the second, and a reader entitled to check either should not have to guess which is which.

Two properties follow, and they are properties of the type system rather than of the prompt. The model’s output is a scalar in a closed interval, so there is no syntactic route by which it can express a weight, relax a ceiling, or admit an ineligible asset. And the solver is the only component that sees Σ , the caps and the turnover penalty, so the trade-off between expected return and risk is made where those quantities exist.

B. Pipeline

Each January the system executes five stages in a fixed order. Stages one to three are deterministic, the model enters only at stage four, and stage five is human.

(i) Dated universe. The universe of record is the exchange’s own historical quotation archive [12], not a

vendor feed. (ii) Available-at-decision fundamentals. Regulatory filings [13], [14] carry both a reference date and a receipt date; only filings received on or before the decision date may enter. (iii) Screen. Assets must clear liquidity, profitability, return on capital and solvency, with operating companies and financial institutions screened on different quantities because their standardised statements are not comparable. (iv) Ranking and allocation. Survivors are ranked and the ranking becomes the confidence vector; allocation is by the solver in the experimental arms and by the score-proportional top- N rule in the published strategy, in both cases within the ceilings and with the residual held in the domestic overnight rate [15]. (v) Human review. The output is a proposal carrying the applied policy, the screen with its rejection reasons, the weights, order sizes with lot rounding and participation in average traded volume, and the estimated cost. No order is transmitted.

The economic hypothesis is fundamental-led rather than language-model-led. Quality and value are the primary accounting dimensions, twelve-month momentum is a market confirmation signal, and liquidity is an execution gate. Banks and operating companies use different accounting quantities. The model does not create this signal: in production architecture it converts approved facts into thesis, risk and review questions. MVO is an independent benchmark in the published record and an allocator only inside the four-arm experiment.

IV. Evaluation infrastructure

A null result is a claim about an instrument as much as about a treatment. This section describes the instrument.

A. Survivorship

A public adjusted-price provider serves the companies that still exist, so a universe built from one silently deletes every company that was delisted, acquired or wound up. Rebuilding from the exchange archive over 2010–2025 yields 4,066 sessions and 514 issuers, of which 166 stop trading before December 2025; 77 of those disappear before 2020, and 58 had already fallen more than 60% from their own peak when they stopped trading. These are precisely the worst returns in the sample.

Delisted tickers are not served by any provider, so corporate actions must be recovered from round, persistent price ratios in the archive itself. We validate the detector against an archived provider event feed on the subset where one exists and publish both sides: precision 88.1% (340 of 386 applied adjustments coincide with a confirmed event) and recall 23.3% (103 of 443 provider events detected). The low recall is a real limitation and is stated as one; the detector is deliberately conservative and prefers not to adjust over adjusting wrongly.

B. Availability gating

Trailing-twelve-month quantities are bridged as annual plus current year-to-date minus comparative year-to-date,

using the year-to-date column only. The interim filing also publishes the isolated quarter, and mixing the two silently corrupts every ratio from the second quarter onward. This is the kind of defect that produces a better backtest and no error message.

C. Holding, cost and tax

The book is bought in January and held. This is worth stating because the natural vectorised expression of a backtest, compounding the inner product of returns and weights, silently rebalances to fixed weights every session at zero cost, which is neither the protocol under evaluation nor anything an investor can execute. Costs are the published exchange fee plus a slippage term growing with the position’s participation in average traded volume. Taxation follows the domestic regime, with gains taxed when a review actually realises them and the final year charged as a full liquidation.

D. Choosing the rule without reading the answer

Our first protocol failed this test, and the failure is instructive. A grid of candidate rules was evaluated, the best was published, and its rank moved from 57th on the declared training criterion to 1st on the holdout, which is only possible if the holdout was read.

The current protocol is nested. For decision year t , the 36 configurations are ranked using only years that closed before t , by Sharpe of the excess return over cash; year t is then evaluated once. Changing configuration is charged a full-turnover rebalance. Ranking requires three closed years, so 2012–2014 is the selection window and 2015–2025 is the evaluation window, giving eleven decisions.

Three quantities are published with every result: the nested outcome, the full-sample winner, and the difference between them. That difference, the hindsight premium, is what an author would gain by publishing the search winner as though it were attainable. On this panel it is 0.65 pp/yr. On a panel with one year less of training it was 4.98 pp/yr, which is a useful calibration of how much the measure moves with sample length.

With 36 trials, the winner’s Sharpe is biased upward by the search itself. The observed excess-return Sharpe is 0.933 against an expected maximum under the null of 0.355, giving a deflated-Sharpe probability of 0.986, above the 95% threshold.

V. Experiment

A. Design

Four arms over thirteen years, same eligible universe, same solver, same caps:

- Named: the model sees company names and returns a bounded score.
- Anonymised: identical, but issuers are identified only by integers.
- Deterministic: control with no model, ranking by the pre-declared factor score.

TABLE I
Four arms, thirteen years, same eligible universe.

Arm	CAGR	Win yrs	Eq. wt.
Named	13.00%	7 of 13	43.8%
Anonymised	12.62%	10 of 13	44.7%
Deterministic	12.67%	8 of 13	52.2%
Monolithic	11.81%	7 of 13	55.0%
Cash (CDI)	9.59%	—	—

TABLE II
The three effects. None survives testing.

Effect	Annualised	p
Contamination (named – anonymised)	+0.38 pp	0.91
Added value (anonymised – deterministic)	−0.05 pp	0.99
Decoupling (anonymised – monolithic)	+0.81 pp	0.78

- Monolithic: the counterfactual in which the model returns weights directly, with no solver and no constraints.

Comparisons are paired by year and tested with a paired t -test.

B. Results

Table II reports three nulls, and each means something different.

No detectable contamination. The named arm did not profit from recognising companies whose fate the model could plausibly know from training. This was the most serious objection to any use of a language model on historical data, and it did not materialise in this configuration. We do not claim contamination is absent in general; we claim it was not detected here, at this model’s training cutoff.

No detectable added value. The model reproduces the deterministic ranking rather than adding judgement. The gap is five hundredths of a percentage point, with a negative sign. This is an uncomfortable finding for the prevailing narrative about language models in investment management, which is a reason to publish it rather than a reason not to.

Decoupling is free. Keeping the model away from the weights did not cost performance. More importantly, the monolithic arm breached no ceiling in thirteen years, a point in its favour that we report because the opposite would have been the convenient finding. What it did produce was arithmetically invalid: weights failing to sum to one in five of thirteen years, ranging from 0.92 to 1.017, and 65 and 83 eligible names silently omitted in 2021 and 2022. The argument for the solver is therefore not that it prevents a limit breach. It is that it returns an allocation requiring no repair, and that repair on a malformed weight vector is itself an unlogged decision.

Thirteen annual observations make these underpowered tests. The honest statement is that no effect was detected, not that none exists.

TABLE III
Retention buffer, nested selection, 2015–2025.

β	CAGR	Excess Sharpe	Carryover	Names lasting 1 yr
0	18.51%	0.597	36%	21
1	18.80%	0.606	42%	16
2	18.70%	0.610	42%	16
3	18.49%	0.593	44%	15
5	18.30%	0.579	44%	15
8	18.30%	0.568	50%	10

C. A robustness result: decision cadence

A reviewer will ask whether an annual decision is simply too slow. We test it by holding the rule, the limits, the universe and the panel fixed and varying only the decision dates. Annual reselection compounds at 18.58% before costs, quarterly at 16.37%, monthly at 13.92%. Paired by calendar year and measured after tax, quarterly trails the annual decision by 1.55 pp/yr ($p = 0.38$) and monthly by 3.27 pp/yr ($p = 0.31$).

The mechanism is not the one usually assumed. The loss sits in the gross return, which falls 4.66 points from annual to monthly before any deduction; execution cost rises by only 0.08 points a year, and modelled tax falls, because each monthly review realises a smaller slice of the book. More frequent reselection degrades the selection before it raises the bill. We note that our tax model understates the monthly arm’s liability, since it charges tax on each period’s own gain rather than on the accumulated gain of the position actually sold; the bias favours the arm that lost anyway.

D. A second robustness result: position persistence

The top- N cut has a property worth measuring: an incumbent that remains excellent but slips one rank is sold in full. Over eleven years the book carried over 36% of its names, and 21 of the 35 names it ever held lasted a single year. We therefore add a retention buffer β : an incumbent is kept while it remains eligible and ranks within $N + \beta$, and the remaining slots are filled from the top of the ranking. $\beta = 0$ is the published rule.

A small buffer helps and a large one hurts, which is the shape one would expect if the factor carries real but decaying information. Paired against the hard cut, $\beta = 1$ gains 0.33 pp/yr ($p = 0.14$) and $\beta = 2$ gains 0.18 pp/yr ($p = 0.12$); neither clears conventional significance on eleven observations. The nested protocol, given β as a further choice, converges on $\beta = 1$ from 2017 and returns 18.61%, a gain of 0.10 pp/yr over the published rule.

We report this as a stability result rather than a return result. The return difference is not established; the change in the book is unambiguous, with carryover rising from 36% to 42% and single-year names falling from 21 to 16. Where that matters is outside the return series: fewer round trips, a longer deferral of realised gains, and a book

TABLE IV
Nested protocol, 2015–2025, net of costs.

Series	CAGR	Win yrs	Max DD
Benevente	17.86%	—	−47.8%
Benevente, after tax	16.03%	—	—
Ibovespa	11.77%	7 of 11	−47.0%
BOVA11 (investable ETF)	11.72%	7 of 11	—
Cash (CDI)	9.61%	6 of 11	0%
Neutral MVO	7.83%	10 of 11	—

an adviser can explain without justifying why last year’s thesis was abandoned.

VI. Published strategy

For completeness, the protocol’s own eleven-year record, net of modelled costs.

The drawdown is not incidental. The nested protocol raised the equity budget to 95% between 2018 and 2021, and the portfolio took essentially the same fall as the index. This is a concentrated book that recovered faster, not a smoothed version of the market. We refer to this canonical annual strategy as Benevente 1.

VII. Experimental event-risk extension

Benevente 2 preserves Benevente 1’s annual holdings and changes only the intrayear equity exposure. It observes the Ibovespa drawdown from a 126-session rolling peak and 20-session realised volatility, both lagged by one close. An alert state caps equity at 50% when drawdown reaches 12% or annualised volatility reaches 40%; a severe state caps it at 35% at 22% and 60%, respectively. Recovery requires ten sessions. The residual earns CDI and each transition pays 10 basis points per unit of turnover. No language model, news text, or future return enters this rule.

The configuration was selected on 2015–2018 and read separately on 2019–2025. Over the full 2015–2025 sample, Benevente 2 produced an 18.45% CAGR and a −28.7% maximum daily drawdown, versus 17.86% and −47.8% for Benevente 1. In 2020, the alert and severe states began on 28 February and 10 March, the minimum equity exposure was 35%, and the annual return was 4.35% versus 1.78% for Benevente 1 and 0.62% for the Ibovespa.

The return claim does not survive a stronger reading. In 2019–2025 the CAGRs are 18.03% and 17.95%, and the paired annual test gives $p = 0.964$. Across 432 sensitivity configurations, every configuration reduces drawdown, but only 9.03% improve both CAGR and drawdown in the evaluation slice. We therefore report robust tail-risk reduction, not added alpha. The experiment includes transition costs but not the tax consequences of intrayear sales. It was also conceived after the Covid-19 episode, so the chronological split cannot remove conceptual hindsight and is not prospective validation.

TABLE V
Defects, all of which had inflated the result.

Defect	Correction
MVO comparator identical to the strategy	Independent implementation
Vendor panel dropped delisted issuers	Universe rebuilt from the exchange archive
Delisting year discarded entirely	Position liquidated into cash
No tax modelled	Domestic tax regime
Flat cost ignoring liquidity	Slippage by participation in volume
Implicit daily rebalancing	Exact buy-and-hold path
Session filter using a global peak	Local rolling peak

VIII. Defects found by auditing our own backtest

Seven defects were found and corrected. Every one inflated the result, which is what one should expect: an error that hurts your own numbers is noticed immediately and an error that helps them is not.

After the final correction the published CAGR fell and the daily drawdown rose from 30.4% to 47.8%. The figures were published as they came out.

IX. Limitations

The evaluation window is also the development window; eleven evaluated years are not eleven years of prospective validation, and the 0.65 pp hindsight premium bounds that problem without eliminating it. The corporate-action detector’s 23.3% recall means distributions and events on delisted names still require reconciliation against primary records before any commercial performance claim. Issuers without provider coverage receive the annual cross-sectional median distribution yield, an approximation whose affected names are listed with the panel. Eleven annual observations are few; the deflated-Sharpe probability corrects the search bias, not the sample size. The language-model experiment uses one model at one training cutoff, and separating contamination from capability requires repeating it across models with different cutoffs. The annual cadence fits the accounting signal and beat quarterly and monthly reselection in this sample, but it is not claimed to be universally optimal. The Benevente 2 extension is retrospective, was designed after Covid-19, and omits intrayear tax; it must be pre-registered and evaluated prospectively before promotion. A separate study should test timestamped news and event-driven or quarterly decisions without contaminating the annual protocol. The factor transformation, eligibility thresholds, configuration grid and annual choices are disclosed in the supplementary artefacts; the source code is not required to understand those transformations.

X. Conclusion

We set out to measure what a language model contributes to portfolio construction when it is denied the ability to set weights. On thirteen years of Brazilian equities, under an architecture where its only output is a bounded score entering a convex program, the answer is: nothing detectable. It does not profit from recognising companies, it does not beat the deterministic factor it was given, and keeping it away from the weights costs nothing. What generates the return is the dated screen, not the ranking method layered on top of it.

We consider this the useful result. An architecture that constrains a model to a bounded, typed output is cheap, gives up nothing measurable, and removes an entire class of failure that the monolithic arm demonstrated concretely by returning weight vectors that did not sum to one. The event-risk extension reinforces the separation: a deterministic, lagged control can materially reduce tail risk without asking the model to infer a crisis from prose. The burden of proof for handing a language model the pen should sit with those proposing it, and meeting that burden requires the deterministic control and the anonymised arm that this protocol makes routine.

Statement on generative AI

A generative-AI assistant was used for language editing and code review. The authors independently verified every calculation, citation and claim and accept full responsibility for the manuscript. The empirical results were generated by deterministic project code, not by the assistant [16].

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